

Treasury-bill return (1%), the TMOM trading rule would suggest that we sit in cash until the return on the risky asset over the past 12 months exceeds the Treasury-bill return over the past 12 months.

Combining MA and MOM Is ROBUST

The examples above outline the moving average and the time series momentum trading rules. As it turns out, there is indeed a formal relationship between MA and TMOM trading rules. We examine the formal relationship between MA and TMOM using basic mathematics. If you are quantitatively challenged, feel free to skip the equations and move on to the figures that follow.

First some definitions:

- Define MA as follows: $MA_t(N) = \frac{x_t + x_{t-1} + \dots + x_{t-N+1}}{N}$ (N period moving average calculated at time t).
- Define TMOM as follows: $TMOM_t(N) = \left(\frac{x_t}{x_{t-N}} - 1 \right) - rf$ (N period excess return calculated at t).

Step 1: Take a difference between the MA at time t and the MA at time $t - 1$, and get the following formula:

$$MA_t(N) - MA_{t-1}(N) = \frac{x_t - x_{t-N}}{N} \quad 1$$

Step 2: Then we break down the TMOM formula as follows:

$$\begin{aligned} TMOM_t(N) &= \left(\frac{x_t}{x_{t-N}} - 1 \right) - rf = \left(\frac{x_t - x_{t-N}}{x_{t-N}} \right) - rf \\ &= \left(\frac{x_t - x_{t-N}}{N} \right) \frac{N}{x_{t-N}} - rf \end{aligned} \quad 2$$

Step 3: We can get the following relationships between formulas (1) and (2):